

SELF-INSTRUCT: Aligning Language Models with Self-Generated Instructions

Yizhong Wang[♣] Yeganeh Kordi[◇] Swaroop Mishra[♡] Alisa Liu[♣]
 Noah A. Smith^{♣+} Daniel Khashabi[♣] Hannaneh Hajishirzi^{♣+}

[♣]University of Washington [◇]Tehran Polytechnic [♡]Arizona State University

[♣]Johns Hopkins University ⁺Allen Institute for AI
 yizhongw@cs.washington.edu

Abstract

Large “instruction-tuned” language models (i.e., finetuned to respond to instructions) have demonstrated a remarkable ability to generalize zero-shot to new tasks. Nevertheless, they depend heavily on human-written instruction data that is often limited in quantity, diversity, and creativity, therefore hindering the generality of the tuned model. We introduce SELF-INSTRUCT, a framework for improving the instruction-following capabilities of pre-trained language models by bootstrapping on their own generations. Our pipeline generates instructions, input, and output samples from a language model, then filters invalid or similar ones before using them to finetune the original model. Applying our method to the vanilla GPT3, we demonstrate a 33% absolute improvement over the original model on SUPER-NATURALINSTRUCTIONS, on par with the performance of InstructGPT₀₀₁,¹ which was trained with private user data and human annotations. For further evaluation, we curate a set of expert-written instructions for novel tasks, and show through human evaluation that tuning GPT3 with SELF-INSTRUCT outperforms using existing public instruction datasets by a large margin, leaving only a 5% absolute gap behind InstructGPT₀₀₁. SELF-INSTRUCT provides an almost annotation-free method for aligning pre-trained language models with instructions, and we release our large synthetic dataset to facilitate future studies on instruction tuning.²

1 Introduction

The recent NLP literature has witnessed a tremendous amount of activity in building models that

¹Unless otherwise specified, our comparisons are with the text-davinci-001 engine. We focus on this engine since it is the closest to our experimental setup: supervised finetuning with human demonstrations. The newer engines are more powerful, though they use more data (e.g., code completion or latest user queries) or algorithms (e.g., PPO) that are difficult to compare with.

²Code and data are available at <https://github.com/yizhongw/self-instruct>

Figure 1: Selected tasks from the generated instruction data using vanilla GPT3. Some texts are reformatted for presentation. See Table 10 for more examples.

can follow natural language instructions (Mishra et al., 2022; Wei et al., 2022; Sanh et al., 2022; Wang et al., 2022; Ouyang et al., 2022; Chung et al., 2022, i.a.). These developments are powered by two key components: large pretrained language models (LM) and human-written instruction data (e.g., PROMPTSOURCE (Bach et al., 2022) and SUPER-NATURALINSTRUCTIONS (Wang et al., 2022, SUPERNI for short)). However, collecting such instruction data is costly and often suffers limited diversity given that most human generations tend to be popular NLP tasks, falling short of covering a

